Supplementary Material for *Large-scale traffic flow reconstruction from heterogeneous mobility sensors*

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This supplementary material documents the meaning of the data layers, model variables, data-assimilation logic, validation metrics and webserver architecture supporting the main manuscript. It deliberately avoids listing repository paths; the focus is on what each component represents, why it is needed, and how it enters the reconstruction workflow. Numerical choices are reported together with their interpretation. Some of them are inherited from the temporal resolution of the observations, some are computational choices that stabilize the simulator, and some are reporting thresholds used to choose clear representative examples for the paper figures.

S1 Data Layers

The reconstruction combines four observational layers and one derived road support. Their roles are distinct and are not interchangeable (Table S1). The mobile-phone layer is the only one that describes regional demand at population scale, but it is too coarse to say which road was used. The road graph provides the missing physical support for route assignment. Coils provide accurate local counts, but only where a sensor exists. Viasat is spatially richer than coils, but it is a probe sample rather than a total-count system. The reconstructed link-flow table is therefore not a direct observation; it is the result of combining these partial views under the explicit model assumptions described below.

Table S1: Meaning and role of the data layers.

Layer	Spatial/temporal support	Meaning in the workflow
Mobile-phone OD activity	Municipality pairs, hourly profiles	Population-scale demand. It defines where trips are generated and how demand changes over the day, but it does not identify the road sequence used by each trip.
Road graph	Directed polylines and nodes	Physical support on which demand is assigned. It stores connectivity, directionality, segment length and turn constraints.
MTS magnetic coils	Fixed points, 15-min counts	Local direct traffic measurements. They anchor the simulation and drive the data assimilation where observed counts are available.
Viasat GPS telemetry	Matched road links, hourly counts	Independent validation layer. Because it samples only part of the vehicle population, it is compared with simulation after estimating link-specific penetration.
Derived link-flow tables	Directed road links, hourly counts	Simulation output after optional data assimilation. These tables are the operational link-level reconstruction used for maps, validation and webserver delivery.

S1.1 Demand Structure

The weekday and holiday OD matrices are highly concentrated. This motivates using the observed spatial demand instead of injecting uniform trips across the region. The total-flow values in Table S2 are the activity weights entering the demand generator after aggregation over the selected January 2020 period. They should be interpreted as relative mobility mass rather than as direct vehicle counts. Positive cells count municipality pairs with non-zero activity. The Gini coefficient and top-share columns quantify concentration: values close to 0.8 and a top-1% share above 0.4 mean that a small number of OD relations dominate the signal. This is why the model uses observed high-demand centres explicitly and treats weakly observed endpoints through random classes instead of forcing unreliable low-count OD pairs into deterministic routes.

Table S2: Structure of the weekday and holiday OD matrices.

Period	Total flow	Positive cells	Gini	Top 10% share	Top 1% share
Weekdays	2,665,186	87,212	0.788	0.708	0.448
Holidays	2,198,516	87,447	0.777	0.690	0.432

The ranking in Table S3 is used only to describe the heterogeneity of municipal activity and to identify the dominant urban centres. The last-ranked municipalities have mean activity equal to one because they are at the residual tail of the aggregated demand table. They are kept in the descriptive statistics, but the simulator does not assume that such low-activity entries carry enough information to define precise origin–destination routes on their own.

Table S3: Municipality activity ranking used to define high-demand centres.

Highest activity			Lowest activity		
Rank	Municipality	Mean points	Rank	Municipality	Mean points
1	Bologna	345,267	519	Montecalvo in Foglia	1
2	Modena	176,166	520	Porto San Giorgio	1
3	Parma	158,440	521	Trento	1
4	Ravenna	147,772	522	Montepulciano	1
5	Rimini	144,696	523	Vescovato	1
6	Reggio nell'Emilia	139,849	524	Tortona	1
7	Ferrara	103,067	525	Aviano	1
8	Piacenza	92,678	526	Malborghetto Valbruna	1
9	Forlì	86,078	527	Bolzano	1
10	Cesena	78,582	528	Grottammare	1

S1.2 Sensor and Telemetry Coverage

Table S4 reports data volume in terms of records, variables and physical interpretation. The 528 municipalities define the areal units of the demand layer. The 292 traffic stations are sparse compared with the road graph, so data assimilation has to be local and directional rather than a global rescaling of the whole network. The 5,760 rows of the 2020 coil table correspond to 60 days at 15-min resolution; each day contributes 96 samples per direction. The Viasat campaigns contain hundreds of thousands of hourly link observations, but the 4,186 matched road identifiers are still only a subset of the regional graph. This justifies using Viasat for temporal-profile validation and not as a complete observation layer.

Table S4: Observed data volumes used by the paper workflow.

Layer	Records	Variables	Interpretation
Municipality activity table	528	5	Ranked source and destination activity over municipalities.
Coil registry	292	5	Geographical catalogue of traffic-count stations.
Coil-to-road matching	292	9	Association between each traffic station and a graph polyline.
February–March 2020 coil table	5,760	496	Fifteen-minute directional traffic counts used for the coil-282 comparison and data-assimilation support.
Viasat 2020 campaign	803,904	10	Hourly GPS link-flow observations over 4,186 matched road identifiers.
Viasat 2021 campaign	703,416	10	Hourly GPS link-flow observations over the same matched road universe.

S2 Simulation Runs and Source Classes

The final validation archive contains sixteen daily simulations, eight in February 2020 and eight in June–July 2021. The two periods were selected because independent Viasat validation campaigns are available for those dates. Viasat is not used to configure or assimilate the simulations; it provides the held-out probe layer against which simulated link profiles are compared. The periods are not intended to span all seasonal variability; they provide two well-defined weeks in which reconstructed link flows and GPS probe flows can be compared consistently.

Every run uses a 10 s integration time step and writes hourly outputs. The 10 s step is short enough to move agents through short links without creating coarse jumps at intersections, while remaining computationally light for regional daily runs. Hourly output is used because both the mobile-phone demand profiles and the validation figures are expressed at hourly scale; it also reduces noise relative to 15-min sensor counts while preserving morning, midday and evening traffic structure. The source count changes by day because the OD generator activates only demand components present in the selected daily schedule.

Table S5: Archived daily simulation configurations, summarized by meaning rather than file name.

Date	Sources	Output support
2020-02-10	758	Hourly directed links
2020-02-11	740	Hourly directed links
2020-02-12	762	Hourly directed links
2020-02-13	736	Hourly directed links
2020-02-14	722	Hourly directed links
2020-02-15	758	Hourly directed links
2020-02-16	980	Hourly directed links
2020-02-17	758	Hourly directed links
2021-06-28	763	Hourly directed links
2021-06-29	744	Hourly directed links
2021-06-30	766	Hourly directed links
2021-07-01	740	Hourly directed links
2021-07-02	726	Hourly directed links
2021-07-03	763	Hourly directed links

Date	Sources	Output support
2021-07-04	983	Hourly directed links
2021-07-05	763	Hourly directed links

The slightly larger source counts on Sundays arise from the way residual holiday mobility is represented: when the observed OD structure is less dominated by commuting pairs, more source components are needed to preserve regional coverage. This does not mean that Sunday traffic is necessarily larger than weekday traffic; it means that the demand generator uses a larger set of active source definitions to represent a more diffuse mobility pattern.

Table S6: Source-class counts across the archived simulations.

Class	Min/day	Max/day	Total	Endpoint logic
Fixed municipality OD	563	804	9,843	Observed source and observed destination.
Fixed province OD	55	78	1,003	Province-level source and destination.
Source-random	50	50	800	Observed source, sampled destination.
Random-destination	50	50	800	Sampled source, observed destination.
Random-random	1	1	16	Both endpoints sampled from the modelled region.

The fixed municipality OD class dominates because it is the part of the demand where both endpoints are directly observed. The two one-sided random classes are kept at 50 sources per day so that residual mobility can enter and leave the main observed centres without overwhelming the empirical OD matrix. The single random–random source is a low-intensity background component: it prevents the model from suppressing trips with neither endpoint in the dominant observed set, but it is deliberately small so that it cannot create an artificial uniform traffic field.

S3 `minimocas` Model Details

`minimocas` is an agent-based road simulator. Vehicles are represented by agents that move on directed polylines, choose a next node at junctions and accumulate link-level counts. The model state is summarized in Table S7. The implementation is intentionally mesoscopic in spirit: agents are individual vehicles, but the objective is not to reproduce lane-changing details or second-by-second car-following trajectories. Instead, the model produces a directional link-by-time flow field that can be locally constrained by fixed counters and validated against held-out probe data.

Table S7: Main vehicle-agent state variables.

State variable	Type/unit	Role
Identifier	integer	Distinguishes each simulated vehicle.
Status	categorical	Tracks whether the vehicle is active, completed, waiting, queued or interacting with transport logic.
Current road link	polyline id	Link occupied by the vehicle at the current time step.
Position on link	m	Longitudinal coordinate along the occupied link.
Signed speed	m s^{-1}	Magnitude gives speed; sign encodes direction.

State variable	Type/unit	Role
Destination	node id	Current graph node pursued by the routing rule.
Previous and next node	node ids	Junction state used to move between links.
Route-deviation parameter	probability	Controls whether the vehicle follows or departs from the precomputed best path.
Distance and time counters	m, s	Accumulate travelled distance, trip time and lifetime.

Best paths are computed on a weighted graph with a Dijkstra-style engine. In the archived validation runs the routing weight reduces to polyline length, while forbidden turns are still excluded. The movement update also contains a density-dependent speed rule:

$$v_p^{\text{target}} = \begin{cases} v_{\text{crit}}, & \rho_p > \rho_{\text{crit}}, \\ (1 - \rho_p/\rho_{\text{crit}})v_p, & \rho_p \le \rho_{\text{crit}}, \end{cases} \qquad v_p^{\text{target}} \ge v_{\text{crit}}. \tag{1}$$

The validation configuration uses $\rho_{\text{crit}} = 3/4.20$ m^{-1} and $v_{\text{crit}} = 10/3.6$ m s^{-1}.

The critical density corresponds to a compact but still interpretable spacing: roughly three vehicles over 4.2 m in the internal density convention used by the simulator. Its role is not to calibrate a detailed fundamental diagram; it prevents unrealistically high speeds when many agents occupy the same link. The critical speed is 10 km h^{-1}, converted to SI units, and acts as a lower operational speed in congested conditions. The rule should therefore be read as a stabilizing traffic-flow constraint rather than as a fitted car-following law.

Table S8: Numerical parameters and switches in the archived validation runs.

Parameter	Value	Role
Integration step	10 s	Numerical simulation step.
Output sampling	3600 s	Temporal resolution of saved link counts.
Source activation	30 s	Interval at which new vehicles are generated.
State grid cell	5000 m	Spatial scale of gridded state output.
Cartographic index cell	10000 m	Spatial index used for graph lookup.
Length routing weight	1.0	Makes routing cost proportional to length.
Speed routing weight	0.0	Disables speed-dependent routing in archived runs.
Initial speed range	15–25 m s^{-1}	Sampling range for generated vehicles.
Random-endpoint time budget	5600 s	Maximum time field used by random source classes.
Unique-link output	active	Saves the directional link counts used in the paper.

The values in Table S8 reflect a compromise between temporal fidelity and reproducibility. A 10 s integration step is small relative to the one-hour saved output and keeps route progression smooth. The 3600 s sampling interval is chosen because the OD demand and Viasat validation are analysed hourly; using a finer output would mostly preserve sensor noise rather than add information to the paper figures. The 30 s source-activation interval spreads injected agents over each hour instead of creating a single burst at the hour boundary. The 5000 m state grid and 10000 m cartographic index are lookup scales used to organize regional graph operations; they are much larger than individual road links, so they do not define the resolution of the final traffic field. Setting the length routing weight to 1.0 and the speed routing weight to 0.0 makes route choice depend on path length in the archived runs. This choice avoids mixing route assignment with uncertain speed calibration when the main objective is to validate link-flow profiles. The 15–25 m s^{-1} initial-speed range covers typical extra-urban speeds before density updates. The 5600 s random-endpoint time budget is a soft bound used when sampling residual trip endpoints: it allows trips longer than a one-hour commute but avoids generating unrealistic cross-region random journeys as the dominant residual class.

S4 Coil-based Data Assimilation

The data-assimilation step adjusts simulated unique-link counts using MTS coil residuals. Directional coil columns are mapped to cartographic `IN`/`OUT` directions. If observed values are missing for the requested interval, seasonal weekday averages are used to fill the signal. This imputation is deliberately conservative: data assimilation needs a reasonable local reference, but missing sensor values should not generate sharp artificial anomalies. The use of weekday averages preserves the typical hourly structure of the corresponding day type while avoiding a global adjustment based on unrelated stations.

For a coil-associated directed link i, let $u_i(h)$ be the unique-link count, $b_i(h)$ the simulated virtual-barrier count and $c_i(h)$ the measured coil count. The data-assimilation step estimates a barrier-to-unique relation and a local relative residual,

$$r_i(h) = \frac{b_i(h)}{\max(u_i(h), 1)}, \qquad \Delta_i(h) = \frac{c_i(h) r_i(h) - b_i(h)}{\max(b_i(h), 1)}. \tag{2}$$

This residual becomes an additive assimilation increment on the local directed link and is propagated to neighbouring links according to simulated turning shares. The ratio $r_i(h)$ is needed because a virtual barrier crossing and a unique-link occupancy count are not the same measurement. The former counts passages through a point-like section, whereas the latter counts distinct agents present on the whole directed polyline during the output interval. Using $r_i(h)$ converts the coil residual into the same scale as the simulated barrier before it is applied to the unique-link table. The denominator thresholds $\max(u_i(h), 1)$ and $\max(b_i(h), 1)$ prevent division by zero in night-time or low-flow hours; they do not add vehicles to the output, but only regularize the residual calculation.

Table S9: Stopping and exclusion rules in recursive data assimilation.

Rule	Implementation
Minimum propagated increment	Stop if the absolute assimilation increment is below one vehicle.
Maximum graph depth	Stop after five propagated links.
Other sensor links	Do not propagate through another coil-associated polyline.
Forbidden turns	Do not propagate through transitions forbidden in the road graph.
Closed or patched roads	Do not propagate through excluded polylines.
Negative counts	Clip negative assimilated counts to zero before delivery.

The one-vehicle stopping rule in Table S9 is a natural discreteness threshold: below one vehicle, the propagated assimilation increment has no operational meaning in an hourly count table. The five-link depth limit keeps data assimilation local. It allows information to pass through a few adjacent roads around a sensor, but prevents a single coil from reshaping a large portion of the regional network. Stopping at other sensor links is also important: once another direct count is reached, that location should be controlled by its own observation rather than by a propagated residual from a neighbouring station.

Table S10: Observed coil 282 data on 10 February 2020.

Direction	Fifteen-minute samples	Daily sum	Peak hourly count	Peak hour
0	96	11,528	1,166	16:00
1	96	10,395	1,143	06:00

Coil 282 is used in the main manuscript because both directions have complete 15-min coverage for the selected day. The daily sums are similar but not identical, indicating directional asymmetry rather than sensor failure. The opposite peak hours, 16:00 in direction 0 and 06:00 in direction 1, are consistent with a road segment that carries different dominant directions in the morning and afternoon. This makes the station useful as a compact example of why direction-aware matching is needed.

S5 Validation Tables

Viasat telemetry is used as an external validation layer for the simulation framework after link-specific penetration scaling. It is not used to generate demand, assimilate counter residuals or tune model parameters. The penetration summary in Table S11 shows that the GPS sampling fraction changes between campaigns. The penetration coefficient η is not assumed to be a universal market share: it is estimated link by link because probe coverage depends on road type, fleet composition and local travel patterns. The Kolmogorov–Smirnov statistic compares the 2020 and 2021 penetration distributions and confirms that the two campaigns should not be pooled without accounting for campaign differences.

Table S11: Full-domain Viasat penetration statistics.

Campaign	Segments	Median $\log_{10}\eta$	Median η [%]	Mean η [%]	KS D
2020	848	-1.396	4.016	6.277	0.109
2021	845	-1.516	3.047	4.686	0.109

Two-sample KS $p = 8.3 \times 10^{-5}$ for the stored 2020–2021 comparison.

The decade classes in Table S12 are used because penetration spans orders of magnitude. Linear bins would hide the distinction between links where GPS represents a few percent of the simulated flow and links where it represents only a few per thousand. The "other" column collects rare links outside the three principal decades and is intentionally not used as a representative validation class.

Table S12: Shares of principal penetration decades.

Campaign	$\eta \sim 10^{-1}$	$\eta \sim 10^{-2}$	$\eta \sim 10^{-3}$	Other
2020	56.60%	35.97%	6.25%	1.18%
2021	48.64%	45.09%	5.80%	0.47%

Table S13 is an aggregate diagnostic, not the main validation claim. High Pearson correlations and matching peak hours show that the simulator and incremented GPS data have compatible daily timing on the best-observed links. However, aggregate agreement can hide local failures; this is why the main text proceeds to subclusters and representative link-level comparisons.

Table S13: High-penetration aggregate-profile fit used to motivate the representative validation class.

Campaign	Segments	Pearson r	L2 distance	Peak hour (simulation/GPS)
2020	480	0.976	0.032	18 / 18
2021	411	0.960	0.035	18 / 18

The global diagnostics in Table S14 summarize the full matched set after excluding links with unusable profiles. The global η is smaller than the median link penetration because high-volume links can dominate the total-flow ratio. Median link correlations and median profile L2 distances are therefore reported together with global quantities: they describe the typical road rather than the volume-weighted aggregate.

The penetration classes in Table S15 explain why the paper focuses on representative high-quality strata. Links in the lowest penetration class have weaker correlations and larger L2 distances, as expected when the probe signal is sparse. Classes with higher median penetration provide more reliable temporal profiles and are therefore better suited for visual validation in the main figures.

Within the representative penetration stratum, the dendrogram in the main text is cut at Ward distance 15,000. This value is not a universal clustering constant; it is the graphical cut that separates the dominant hourly-profile families without fragmenting the validation set into many single-road groups. At lower cuts, several branches represent minor variations of the same daily shape and are not useful for a concise paper figure. At higher cuts, distinct morning/evening profile families begin to merge. The adopted cut therefore provides three interpretable subclusters, which is also the number of representative roads that can be shown clearly in the validation panel.

Table S14: Global validation diagnostics from archived validation tables.

Campaign	Common links	Valid links	Global η	Pearson raw	Median link r	Median profile L2
2020	930	827	0.00937	0.939	0.826	0.0859
2021	927	828	0.00778	0.908	0.766	0.0933

Table S15: Validation quality by penetration class in 2020.

Class	Links	Median η	Median Pearson	Median L2
-4	22	0.000232	0.437	0.290
-3	274	0.001578	0.662	0.134
-2	514	0.008860	0.868	0.073
-1	17	0.037783	0.913	0.061

The representative-link selection in Table S17 balances three constraints. First, the three roads must belong to different temporal subclusters so that the paper does not validate the same daily profile three times. Second, the mean normalized-profile error must remain below 0.80%, which keeps the selected examples within a clear sub-percent discrepancy range while still allowing enough candidates. Third, the selected roads must be at least 40 km apart whenever possible, so the map communicates spatial coverage rather than three neighbouring links. Polyline 104671 is kept fixed because its urban position near Forlimpopoli gives it a direct interpretation; the other two are then selected to satisfy the error and separation criteria.

Table S16: Candidate-filtering logic for the representative validation links.

Selection stage	Links	Purpose
Common simulated–Viasat links in 2020	930	Initial matched comparison set.
Valid hourly profiles	827	Exclude unusable or degenerate profiles.
Representative high-penetration class	17	Candidate set for temporal subclustering.
Displayed representative links	3	One illustrative link per temporal subcluster.

S6 Aggregate Road-safety Context

Figure S1 summarizes the aggregate accident context used to interpret the exposure-based safety analyses presented in the main manuscript. The objective of this figure is not to identify geolocated crash hotspots, but to characterize the temporal and infrastructural structure of the regional accident dataset before combining it with reconstructed traffic exposure.

Panel **a** reports the total number of recorded accidents and deaths between 2022 and 2024. The total accident count remains relatively stable across years, while fatalities show a moderate decrease over the same period. This indicates that the exposure analysis is performed on a temporally consistent regional dataset without large interannual discontinuities.

Panel **b** shows the distribution of accidents across road environments. Urban roads account for the majority of events in all years, whereas highways and ramps contribute only a small fraction of the total recorded accidents. This imbalance is important when interpreting exposure-normalized hazard indicators because high-flow extra-urban corridors may carry substantial traffic volumes while still representing a limited share of total accidents.

Panel **c** compares the seasonal accident distribution with the reconstructed simulated exposure field. The observed accident share increases during summer and autumn, while the simulated weekday exposure is more concentrated in winter and spring. This mismatch suggests that accident occurrence cannot be explained solely by reconstructed traffic intensity and may also depend on seasonal behavioural, touristic

Table S17: Representative polylines shown in Fig. 6 of the main manuscript.

Subcluster	Polyline	Mean error [%]	Selection note
1	104671	0.528	Fixed urban representative near Forlimpopoli.
2	187204	0.761	Selected under error and separation constraints.
3	132443	0.611	Selected under error and separation constraints.

Minimum pairwise separation among the three selected links: 41.9 km.

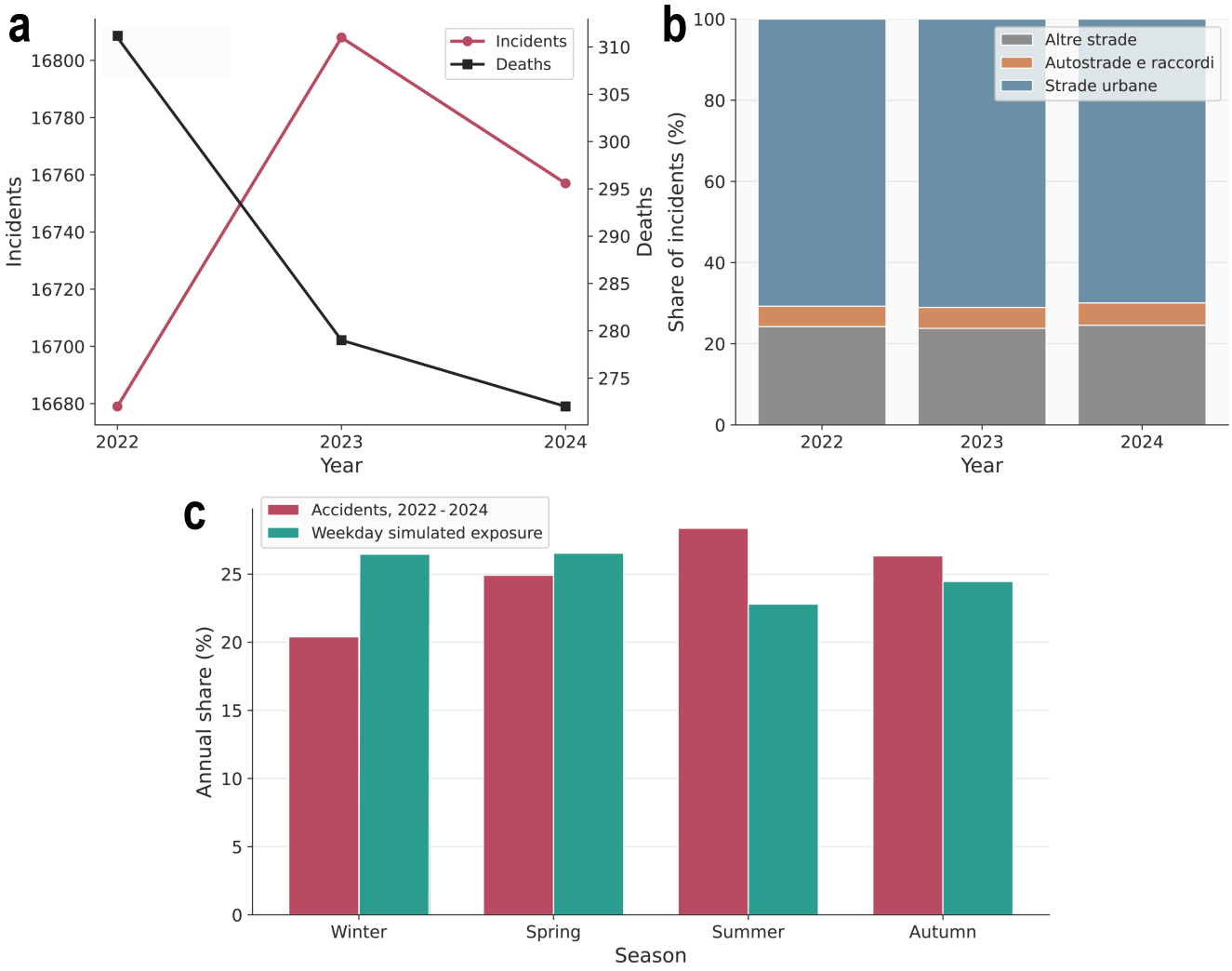


Figure S1: Supplementary overview of the accident data used for the exposure-normalized road-safety screening. The panels summarize aggregate accident totals, road-environment composition and seasonal shares. These descriptors are used to contextualize the main-text exposure analysis; they are not interpreted as geolocated single-event hotspots.

or environmental factors.

The aggregate descriptors shown in Fig. S1 therefore provide the statistical context for the hazard-weighted exposure analyses discussed in the main text and motivate the use of exposure-normalized indicators rather than raw accident counts alone.

S7 Webserver Architecture

The project also implements the reconstruction as an operational web system. The webserver does not change the modelling equations; it wraps the same configuration generator, simulator and data-assimilation procedure into a service that users can launch from a browser. This distinction matters: the webserver is not an additional calibration method, but an operational layer that makes the validated workflow repeatable for users who need maps, tables and downloadable outputs rather than command-line execution.

Table S18: Logical components of the webserver.

Component	Function
Django graphical interface	Authenticated browser interface. It allows a user to submit a simulation interval, inspect submitted runs, download results and open interactive maps.
FastAPI simulation service	Programmatic backend that receives simulation requests, generates a run configuration, launches the simulator in a separate process and exposes status/results endpoints.
Configuration generator	Converts requested dates, coil availability and OD activity into the simulation input used by the backend.
Simulation worker	Runs `minimocas` from the generated configuration and writes status updates while the model advances.
Data-assimilation module	Applies the coil-driven data-assimilation step before results are returned to the user. A special mode returns provincial min–max intervals for institutional users.
Interactive map layer	Renders cartography and simulated fluxes by colouring road polylines by quantile or by selected result values.
MTS monitoring views	Display coil time series, station maps, rankings, clusters and local anomaly diagnostics.
Scheduled data updater	Downloads and converts the public MTS traffic records into the standardized coil tables consumed by the simulator.
Reverse proxy and containers	Provide a deployable service layout with separate GUI, simulation API, scheduler and web proxy containers.

The separation between interface, simulation service and worker process keeps long simulations from blocking the browser. The graphical interface stores metadata and displays results; the backend validates requests and coordinates simulation and data assimilation; the worker performs the computationally expensive network loading. The scheduled updater is included because the counter layer is not static: new MTS observations must be converted into the standardized format before they can be used for monitoring or data assimilation.

The workflow in Table S19 mirrors the scientific pipeline of the main manuscript. User input defines the temporal window, the backend generates a model configuration, the simulator produces raw flows, the data-assimilation step applies coil information, and the final link-flow table is stored for visualization and export. The asynchronous status query is practical rather than methodological: a regional simulation can take longer than a normal HTTP request, so the interface polls progress instead of waiting for a blocking response.

The conceptual API in Table S20 exposes only operations needed by the workflow. The road-graph export allows the browser to draw the same network used by the simulator. The result query applies data

Table S19: End-user workflow implemented by the dashboard.

Step	Action	Result
1	Choose interval	The browser submits start time, stop time and output sampling resolution.
2	Create simulation	The backend validates the request, generates a unique run identifier and builds the configuration.
3	Run asynchronously	The simulation starts in a separate process while the interface polls progress.
4	Assimilate results	After completion, the backend applies the default coil-based data-assimilation step or an institutional min–max adjustment.
5	Persist metadata	The interface stores run identifier, user, status, runtime, completion and result location in the application database.
6	Visualize and export	Users can inspect time profiles, view flux maps and download data-assimilated link-flow tables.

Table S20: Conceptual API exposed by the simulation backend.

Operation	Meaning
Welcome/status handshake	Returns service availability and version.
Simulation submission	Accepts a time interval and creates a simulation task.
Road-graph export	Returns the road graph geometry used by the interface.
Progress query	Reports whether a submitted run is running, complete, invalid or in error.
Result query	When requested after completion, applies data assimilation and returns the reconstructed link-flow table as JSON or CSV.

assimilation at delivery time so the downloaded table and visualized map correspond to the data-assimilated reconstruction, not merely to the raw simulation.

S8 Reproducibility Notes

The tables above were derived from stored analysis outputs and source-code inspection. No new simulation run is required to reproduce the reported counts, validation summaries or dashboard architecture. The operational dashboard uses the same simulation and data-assimilation logic described in Sections S3 and S4; it is therefore an interface to the reconstruction workflow rather than a separate model. Values such as the 10 s integration step, hourly output, five-link data-assimilation depth, 0.80% representative-error threshold and 40 km separation threshold should therefore be interpreted in context. The first two define the numerical and observational time scales, the data-assimilation depth controls the spatial locality of sensor information, and the last two are transparent criteria for choosing readable paper examples rather than hidden calibration parameters.